

Bootstrapping Matrix SYK Models

Abstract. “Matrix SYK” models are purely fermionic $U(N)$ matrix quantum mechanics with a cubic supercharge, $Q = \sum C_{abc} \text{Tr} \Psi^a \Psi^b \Psi^c$ and $H = \{Q, \bar{Q}\}$. The three-flavour version proposed by Chen is the simplest candidate for a matrix model whose BPS sector is *fortuitous* and *R-charge concentrated* — the two properties that, in $\mathcal{N} = 2$ SYK, give rise to a super-Schwarzian low-energy description and to black-hole-like counting of BPS microstates. The model has a finite Hilbert space, so exact diagonalisation settles everything at $N = 2$ and almost nothing beyond, and its interesting ground states have no classical large- N saddle. We are developing the quantum-mechanical matrix bootstrap as the tool for $N > 2$. So far we have: a sector-resolved bootstrap that is exact for the single-matrix model up to $N = 4$ and for the three-matrix model at $N = 2$ in every sector we have tested; a reformulation in terms of trace words whose cost does not grow with N , which establishes $E_0(k; N) = 16N^3 - (15 + 38k)N$ for the low sectors $k \leq N - 1$ and gives the first rigorous bound in an interacting sector at $N = 3$; an exact refined Witten index that reproduces the Turiaci–Witten $\cos(\pi k/3)$ BPS profile and shows e^{cN^2} multiplicities per irrep; and a “BPS-exclusion” bootstrap that rules out BPS states sector by sector in seconds at any N , currently reaching half-way to the BPS window. Index and exclusion together are a concrete route to establishing R-charge concentration at $N = 3$ and 4.

1 Introduction

The models. The degrees of freedom are complex fermionic $N \times N$ matrices Ψ_{ij}^a , with $a = 1, \dots, p$ a flavour index, obeying $\{\Psi_{ij}^a, \bar{\Psi}_{kl}^b\} = \delta^{ab} \delta_{il} \delta_{jk}$; $U(N)$ acts by conjugation. Everything is generated by a nilpotent cubic supercharge:

$$Q = \sum_{a,b,c} C_{abc} \text{Tr}[\Psi^a \Psi^b \Psi^c], \quad H = \{Q, \bar{Q}\} \geq 0, \quad [N_\Psi, Q] = 3Q, \quad N_\Psi = \text{Tr} \Psi^a \bar{\Psi}^a. \quad (1)$$

The fermion number N_Ψ plays the role of the R-charge, and Q raises it by three. For a single flavour, $Q = \text{Tr} \Psi^3$, this is Chen’s model [1], and it is exactly solvable: the Hamiltonian is a function of the gauge Casimir, $H = 3N(N^2 - 1) - 9\hat{C}_2$, and the BPS states (those with $E = 0$) occupy a window of $N + 1$ consecutive R-charges, $N_\Psi \in [\frac{N(N-1)}{2}, \frac{N(N+1)}{2}]$. With three flavours and Chen’s couplings C_{abc} (the cyclic symmetrisation of the indicator function of $a \leq b \leq c$) the Casimir structure is lost. We derived the exact single-trace form of H for this case — a nine-term expression that also contains a $\text{Tr} \Psi^c \text{Tr} \bar{\Psi}^d$ piece coupling the trace modes — and verified that it is *not* a function of Casimirs. This “three-matrix model” is our target.

Why it is interesting. In $\mathcal{N} = 2$ SYK [5] the BPS states are concentrated in a window of $O(1)$ R-charges around half filling, with counts following $\cos(\pi k/3)$. This is the fingerprint of the $\mathcal{N} = 2$ super-Schwarzian [4] and, in the fortuity picture of Chang and Lin [3, 2], of a BPS sector built from black-hole-like “fortuitous” states: states with e^{cN^2} multiplicities that no large- N saddle sees. The single-matrix model fails this test in two ways — its window grows with N , and its BPS states form one enormous irrep with $O(1)$ multiplicity. The three-matrix model at $N = 2$, by contrast, has 972 BPS states in exactly three sectors, $N_\Psi \in \{5, 6, 7\}$, with counts 243:486:243: the 1:2:1 SYK pattern. What happens for $N \geq 3$ is unknown.

Research questions.

- Q1. Does the three-matrix model exhibit R-charge concentration for $N \geq 3$: an $O(1)$ window, $\cos(\pi k/3)$ counts, and e^{cN^2} multiplicities per irrep?
- Q2. Is it chaotic? The natural diagnostic is the Altland–Zirnbauer statistics of the singular values of Q between adjacent sectors.
- Q3. Is three the “magic number” — the minimal flavour structure that breaks the Casimir shortcut and introduces genuine R-charge dynamics and chaos?
- Q4. What are the energies of the near-BPS sectors, and how do they scale with N ? The super-Schwarzian predicts gaps growing like q^2 with the distance from the window.

What makes these models distinctive, and why a bootstrap. There are no bosons. The Hilbert space is the 2^{pN^2} -dimensional Fock space, every operator algebra truncates, and in principle everything is finite-dimensional linear algebra. In practice, 2^{27} at $N = 3$ already puts the full spectrum out of reach (only sectors with a few particles, $k \leq 4$, can still be diagonalised exactly), and 2^{48} at $N = 4$ ends exact diagonalisation for good. At the same time, the states we care about have no classical large- N description: fortuitous states carry maximal Casimir and are as far from gauge singlets as possible, so the standard planar bootstrap — which assumes factorisation — is

blind to exactly this physics. A *rigorous, finite- N* bootstrap, producing lower bounds on sector ground energies and infeasibility certificates for BPS states, is therefore the natural tool here, and, we will argue, a promising one.

2 Methodology

The sector-resolved bootstrap at finite N . The basic object is the ground energy $E_0(k)$ of the sector with $N_\Psi = k$. Rather than a state, we optimise over linear functionals $\phi(O) = \text{Tr}[\rho_k P_k O P_k]$, where ρ_k is a Hermitian matrix on the k -block that is *not* assumed positive. The optimisation problem is

$$\boxed{\begin{array}{ll} \min_{\rho_k = \rho_k^\dagger} \phi(H) & \text{subject to} \\ \phi(\mathbf{1}) = 1, & [\phi(\text{Tr} w_a^\dagger w_b)]_{ab} \succeq 0, \quad [\phi((\text{Tr} w_a)^\dagger \text{Tr} w_b)]_{ab} \succeq 0, \quad \phi([H, X]) = 0, \quad \phi((N_\Psi - k)X) = 0. \end{array}} \quad (2)$$

Here w runs over *open* (adjoint-valued) words in the letters $\Psi^a, \bar{\Psi}^a$ up to some length L , which we call the level. The two matrix inequalities are positivity of the Gram matrix of the states $w_{ij}|\psi\rangle$ in its adjoint and singlet channels; the equalities are the equations of motion and the statement that the functional lives in sector k , imposed for every operator X that appears in the cones. Every constraint is satisfied by the true sector ground state, so the minimum is a rigorous lower bound on $E_0(k)$, and it can only improve as L grows. The dual problem produces a sum-of-squares certificate $H - E \succeq 0$ on the sector — the direct analogue of the functional in the conformal bootstrap.

Two ingredients turned out to matter most. The first is the sector projection itself: charged operators have $\phi = 0$, and commutators act within the block. The second is symmetry: ρ_k may be taken invariant under $SU(N)_{\text{gauge}} \times \mathbb{Z}_3$ at no cost (group averaging), and once it is, the singlet/adjoint decomposition of the Gram matrix introduced by Cho et al. [8] becomes available, $\phi(\text{Tr} w_a^\dagger w_b) - \frac{1}{N} \phi(\text{Tr} w_a^\dagger \text{Tr} w_b) \succeq 0$. We implement the invariance through an isotypic decomposition of the block, $\rho_k = \oplus_R \sigma_R \otimes \mathbf{1}$, which leaves $D = \sum_R m_R^2$ parameters instead of d_k^2 — at most 90 unknowns at $N = 2, k = 2$, and 666 at $k = 3$. The cost is nevertheless set by the block dimension d_k (at most 924 at $N = 2$, but up to 2×10^7 at $N = 3$), so this version of the bootstrap is a validation tool rather than the route to large N .

An exact finite- N formulation in terms of traces. To go beyond $N = 2$ we re-express the same constraints as identities among expectation values of *trace words*, $\phi(\text{Tr} w_1 \cdots \text{Tr} w_m)$, with N entering only as a numerical coefficient. Only two facts about the fermionic matrix algebra are needed. Moving a letter cyclically around a trace preserves the matrix structure and produces double traces,

$$\text{Tr}[\ell_1 \ell_2 \cdots \ell_L] = (-1)^{L-1} \text{Tr}[\ell_2 \cdots \ell_L \ell_1] + \sum_m (-1)^m \delta_{\ell_1 \bar{\ell}_m} \text{Tr}[\ell_2 \cdots \ell_{m-1}] \text{Tr}[\ell_{m+1} \cdots \ell_L], \quad (3)$$

whereas swapping two adjacent letters inside a trace does not preserve it — so no normal ordering is ever performed. The important point is what we do *not* do: we impose no large- N factorisation. Multi-trace expectation values are kept as independent unknowns, so every bound is exact at the chosen integer N , and the planar limit — which would wash out the $1/N$ -suppressed concentration physics — is never taken. The sector condition becomes the linear family $\phi((N_\Psi - k)X) = 0$, and the finite- N trace relations (the vanishing antisymmetriser over $N + 1$ indices) are optional rows that turn out to matter only at $N = 2$. The resulting semidefinite programme is built once, evaluated at any N one likes, and handed to an interior-point (Clarabel) or first-order (SCS) solver. Its size is set by the level, not by N and not by k .

Complementary tools. Three smaller pieces round out the toolkit. A *few-body exact diagonalisation* works directly in the sector $N_\Psi = k$, on the basis of k -subsets of the $3N^2$ fermionic modes, without ever forming the Fock space; the sector dimension is $\binom{3N^2}{k}$ (for instance 351 for $N = 3, k = 2$ and 17550 for $N = 3, k = 4$), so it gives exact ground energies for a few particles at moderate N — precisely where the trace bootstrap operates — and nothing near half filling. An exact *refined Witten index*, resolved by $N_\Psi \bmod 3$, $U(N)$ irrep and $\mathbb{Z}_3$ flavour charge, is discussed in Sec. 3. And the BPS-exclusion variant of (2) is described there and in Sec. 4.

3 Results

Single-matrix model, where the answer is known. At level 2 the sector bootstrap reproduces the exact Casimir ladder in all ten sectors at $N = 3$ (72, 45, 18, 0, 0, 0, 0, 18, 45, 72) and in 15 of the 17 sectors at $N = 4$. The two exceptions at $N = 4$ are the sectors adjacent to the BPS window, $k = 5$ and $k = 11$, whose true energy is 18; there the bound climbs with level, 0, 0, 11.4, 14.7 at levels 2 to 5 (Fig. 1a). An ablation at $N = 3$ shows what each ingredient does: the sector restriction fixes $k = 1$, the adjoint channel fixes $k = 2$, the supercharges in the singlet cone supply the SUSY floor $E \geq 0$, and ground-state positivity is not needed.

sector k	0	1	2	3	4	5	6	7
exact $E_0(k)$ (full ED, 2^{12} states)	98	22	5.16536	1.22706	0.08796	0	0	0
best bootstrap lower bound	98	22	5.16536	1.22706	0	0	0	0
level at which the bound is exact	2	2	3	4	> 4	2	2	2

Table 1: Three-matrix model at $N = 2$. Sectors $k \geq 8$ follow by particle-hole symmetry, $k \rightarrow 12 - k$. The level-3 and level-4 bounds use the symmetry-reduced formulation with the singlet/adjoint split.

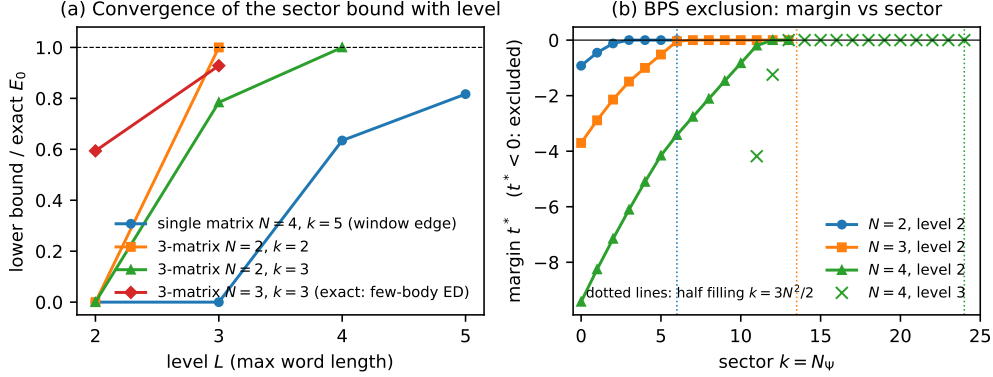


Figure 1: (a) Sector lower bound divided by the exact ground energy (full ED at $N = 2$, few-body ED at $N = 3$) as a function of level, for four representative sectors: each lifted sector becomes exact one level after leaving the SUSY floor. (b) BPS-exclusion margin t^* as a function of the sector: a negative margin certifies that the sector contains no BPS state at that level. At level 2, which takes seconds per sector, BPS states are excluded for $k \leq 2, 6, 11$ at $N = 2, 3, 4$; level 3 adds $k = 12$ at $N = 4$. The dotted lines mark half filling, where the BPS window sits.

Three-matrix model at $N = 2$. Table 1 summarises the sector ground energies. Every sector with a positive gap, except the almost-BPS $k = 4$, is reproduced *exactly*: $k = 2$ at level 3 and $k = 3$ at level 4, while the BPS window $\{5, 6, 7\}$ is exact automatically through the SUSY floor. Two lessons carry over to everything that follows: a lifted sector becomes tight one level after it leaves the floor, and the singlet/adjoint split is what closes the gap (with the plain adjoint channel the $k = 2$ bound stalls at 4.55). The bootstrap also produces *allowed regions*. Fixing $\phi(H) = E$ and extremising an observable maps out an island in the $(E, \langle \hat{C}_2 \rangle)$ plane that encloses the exact spectrum (Fig. 2); for the single-matrix model the island degenerates to the exact Casimir line. With the eigenstate constraints $\phi(OH) = E \phi(O)$ added, the level-2 problem becomes feasible *only at the exact eigenvalues* (5.16536, 12, 13.4907, each to 10^{-8}) — an archipelago rather than an island.

Beyond $N = 2$ with the trace formulation. A single constraint set, evaluated at $N = 2, 3, 4, 5, 10, 30, 100$, reproduces the analytic anchors $E_0(0) = 16N^3 - 15N$ and $E_0(1) = 16N^3 - 53N$ to eight digits. The new results are collected in Table 2. The level-2 bound is *tight* for $k = 2$ at every $N \geq 3$ and for $k = 3$ at every $N \geq 4$, and in all these cases it equals $16N^3 - (15 + 38k)N$; few-body ED confirms the values wherever it can ($N \leq 5$). Since a simple variational state gives the same energy from above, this establishes

$$E_0(k; N) = 16N^3 - (15 + 38k)N \quad \text{for } k \leq N - 1 \text{ (all cases tested)}. \quad (4)$$

	$N = 3$	$N = 4$	$N = 5$	$N = 10$	$N = 100$
$E_0(2; N)$, bootstrap (level 2)	159.000	660.000	1545.000	15090.000	1.59909×10^7
$E_0(2; N)$, few-body ED	159	660	1545	—	—
$E_0(3; N)$, bootstrap (level 2)	45 (level 3: 70.35)	508.000	1355.000	14710.000	—
$E_0(3; N)$, few-body ED	75.7917	508	—	—	—

Table 2: Three-matrix model, low sectors at larger N : rigorous lower bounds from the trace bootstrap (one constraint set evaluated at each N ; SCS at $N = 100$) next to the few-body exact diagonalisation, which is available only where $\binom{3N^2}{k}$ is small (dashes: out of reach). All tight entries equal $16N^3 - (15 + 38k)N$; at $N = 10$ and 100 the bootstrap values coincide with that formula. The one non-tight entry is $k = 3$ at $N = 3$: this is the first *interacting* sector ($k = N$), and at level 2 the bound is only the non-interacting value $16N^3 - 129N = 45$ — the level-2 words cannot see the interaction; level 3 raises the bound to 70.35 against the exact 75.79.

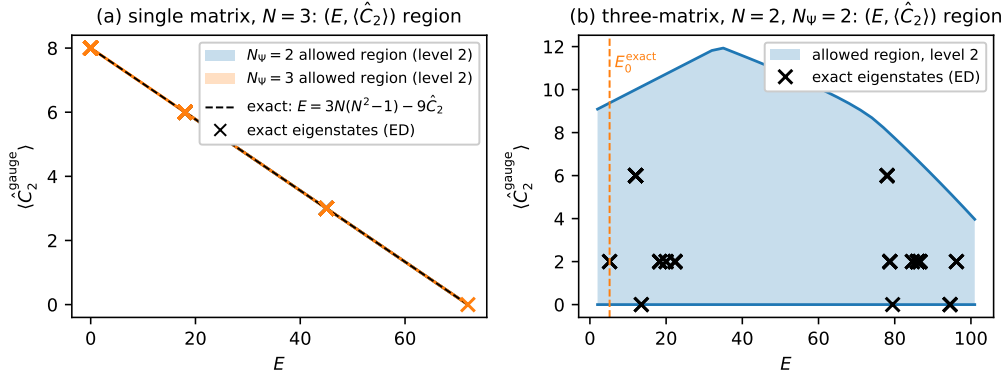


Figure 2: Allowed regions in the (energy, gauge-Casimir) plane from the level-2 sector bootstrap, obtained by fixing $\phi(H) = E$ and extremising $\phi(\hat{C}_2)$, compared with the exact eigenstates (crosses). (a) Single-matrix model, $N = 3$, sectors $N_\Psi = 2, 3$: the region has zero width and coincides with the exact line $E = 3N(N^2 - 1) - 9\hat{C}_2$, because that operator identity lies in the span of the constraints. (b) Three-matrix model, $N = 2$, $N_\Psi = 2$: a genuine two-dimensional island, rigorous and already tight at the ground-state edge.

In words: the k -particle sector is *free* as long as $k \leq N - 1$. The lowest one-particle multiplet is an $SU(N)$ adjoint, and k antisymmetrised copies of it, in the component of $\wedge^k \mathbf{adj}$ with the largest Casimir, are annihilated by the quartic interaction. We checked this explicitly at $N = 3$: $\wedge^2 \mathbf{8} = \mathbf{10} \oplus \overline{\mathbf{10}} \oplus \mathbf{8}$, and the 20-dimensional part is free while the adjoint is lifted by +100. The first sector where interactions act is therefore $k = N$. There the bootstrap gives the first rigorous bound in an interacting sector beyond $N = 2$: $E_0(3; 3) \geq 70.35$ at level 3, against the exact value 75.79 from few-body ED in the 2925-dimensional sector (level 2 gives the free value 45).

An analytic complement: the refined index. Because Q raises N_Ψ by three, the sectors organise into three strands, $\mathcal{H}_c \rightarrow \mathcal{H}_{c+3} \rightarrow \mathcal{H}_{c+6} \rightarrow \dots$ with $c = k \bmod 3$, each further split by $U(N)$ irrep λ and $\mathbb{Z}_3$ flavour charge ω . The Euler characteristic of a strand — the alternating sum of its sector dimensions — is a purely kinematic quantity, and its modulus is a lower bound on the number of BPS multiplets in that strand, with equality exactly when the BPS states of the strand sit at a single R-charge. This “index saturation” is the same statement as R-charge concentration [2]. Using $\mathbb{Z}_3$ -Fourier flavour letters, the Fock character $\prod_m \prod_{i,j} (1 + tz^m u_i/u_j)$ gives every such index exactly. At $N = 2$ the index equals the ED count in all 36 strands. The totals per strand have the closed form

$$I_c(N) = \frac{2}{3} 3^{3N^2/2} \cos\left(\frac{2\pi c}{3} + \frac{\pi N^2}{2}\right), \quad (5)$$

which is *exactly* the Turiaci–Witten profile $\cos(\pi(k - k_*)/3)$ about half filling: 2:1:1 for even N and 0:1:1 for odd N . At $N = 3$ the index vanishes in every $k \equiv 0$ strand. The multiplicities per irrep grow like $e^{0.75N^2}$ (18, 891, 3.5×10^5 at $N = 2, 3, 4$) and include gauge singlets (6, 81, 1944). The single-matrix model, put through the same computation, has a single irrep per strand with multiplicity at most 9 and no singlets at all — the fortuitous/black-hole-like dichotomy of Chang and Lin, now made quantitative.

BPS exclusion (preliminary). A BPS state satisfies $Q\psi = \bar{Q}\psi = 0$, and therefore the linear relations $\phi(XQ) = \phi(QX) = \phi(X\bar{Q}) = \phi(\bar{Q}X) = 0$ for every operator X . Adding these rows to (2) and maximising a feasibility margin turns the bootstrap into a certificate of *absence*: a negative margin means the sector contains no BPS state. At $N = 2$ this excludes $k \leq 2$ at level 2 and $k \leq 3$ at level 3 — one level earlier than the energy bound each time. At $N = 3$ and $N = 4$, level 2 already excludes $k \leq 6$ and $k \leq 11$ in seconds, and level 3 adds $k = 12$ at $N = 4$ (Fig. 1b). Since the index fixes the BPS count of each strand, emptying every R-charge but one by exclusion would determine the BPS spectrum completely.

4 Outlook

- **Concentration at $N = 3, 4$ from index and exclusion together.** The index says the $k \equiv 0$ strands are empty at $N = 3$ and fixes the counts of the others; exclusion currently empties $k \leq 6$ at $N = 3$ and $k \leq 12$ at $N = 4$, about half-way to the window. Two natural next steps: irrep-resolved exclusion (the quadratic Casimir is a trace polynomial, hence a linear constraint), which needs to target only the irreps that actually carry index; and exact cohomology ranks at $N = 3$, computed sparsely over a prime field, as an independent check.
- **Near-BPS energies as a function of N .** The level-3 trace engine gives rigorous, N -dependent bounds on

the interaction energy of the first interacting sector $k = N$ in minutes per N ; calibrated against few-body ED at $N = 4$ (dimension 1.9×10^5), this is a direct test of the super-Schwarzian scaling.

- **Sparse diagonalisation at $N = 3$.** The sectors near the window have dimensions up to 2×10^7 , feasible with a matrix-free Lanczos implementation: energies, exact BPS counts, and the singular-value statistics of Q_k for the chaos question.
- **Analytic work.** A proof of the free-sector formula (a short Casimir argument), large- N asymptotics of the per-irrep index from a saddle point of the character integral, and the traceless $su(N)$ and two-flavour variants, which bear on the “magic number” question.
- **Solvers.** The level-4 problem is assembled but not yet solved to tolerance (see the appendix). The dual, SDPA/SDPB-style form after eliminating the equalities, or the reversal $\mathbb{Z}_2$ symmetry that halves every cone, would bring it within reach of interior-point methods.

A Numerical implementation

- **Finite- N engine** (`sector_bootstrap.py`). Open words are built as blocks between sectors, so the 2^{pN^2} -dimensional space is never formed. An isotypic reducer maps every operator to its $D = \sum_R m_R^2$ invariant components as soon as it is built, and the spans of the touched operators and of the equations of motion are accumulated as streaming $2D \times 2D$ Gram matrices. A diagnostic that evaluates every constraint on the exact, multiplet-averaged ground state caught a round-off Hermitian part being normalised into a spurious constraint; all such cutoffs are now relative to $\|X\|$.
- **Trace algebra** (`trace_algebra.py`). One primitive — swapping two adjacent letters in a *routed term* (letters in operator order plus a routing permutation; the anticommutator branch rewires the routing, closed index loops give factors of N) — generates canonical cyclic forms, products, daggers, commutators with H , self-symmetry reductions such as $\text{Tr}[\ell\ell] = 0$, and the finite- N relations. Coefficients are polynomials in N . Every identity is checked against explicit operators at $N = 2, 3$ (a thousand random terms to 10^{-15} ; the algebra recovers $\{Q, \bar{Q}\} = H$ on its own).
- **Assembly and conditioning** (`trace_bootstrap.py`). Variables are generated by closure; reality is imposed at the variable level (halving the unknowns); rows are unit-normalised; the unknowns carry a 't Hooft scaling $N^{\sum(1+L_i/2)}$ and the objective is divided by N^3 , without which nothing works beyond $N \approx 30$. Solvers are called directly. Settings that matter: Clarabel with chordal decomposition off and equilibration on; SCS with $\rho_x = 10^{-3}$, the Apple-Accelerate LDL backend and fixed scaling. A parallel fixed- N build produces the level-4 problem (51k monomials, cones up to dimension 168) in 25 seconds.
- **Cost and limits.** Levels 2 and 3 take seconds to minutes at any N and at most 5 GB. Level 4 is solver-limited: interior-point codes keep a dense block per cone that grows like the square of its vectorised dimension (about 25 GB for a 168-dimensional complex cone), and the first-order solver stalls at 10^{-2} residuals. Every run is preceded by a memory estimate and guarded by a watchdog.
- **Few-body ED and index** (`fewbody.py`, `bps_index.py`). Trace operators act on k -subset basis states by explicit index sums. The index is exact integer arithmetic: the Fock generating function is built by integer shifts on the $SU(N)$ torus, and irrep multiplicities follow from the Weyl alternating sum; seconds for $N \leq 4$.

References

- [1] Y. Chen, *Fortuity with a Single Matrix*, arXiv:2511.00790.
- [2] C.-M. Chang, Y. Chen, B. S. Sia, Z. Yang, *Fortuity in SYK models*, arXiv:2412.06902.
- [3] C.-M. Chang, Y.-H. Lin, *Holographic covering and the fortuity of black holes*, arXiv:2402.10129.
- [4] G. J. Turiaci, E. Witten, $\mathcal{N} = 2$ JT supergravity and matrix models, arXiv:2305.19438.
- [5] W. Fu, D. Gaiotto, J. Maldacena, S. Sachdev, *Supersymmetric SYK models*, arXiv:1610.08917.
- [6] X. Han, S. A. Hartnoll, J. Kruthoff, *Bootstrapping matrix quantum mechanics*, arXiv:2004.10212.
- [7] H. W. Lin, Z. Zheng, *High-precision bootstrap of multimatrix quantum mechanics*, arXiv:2507.21007.
- [8] M. Cho, B. Gabai, J. Sandor, X. Yin, *Thermal bootstrap of matrix quantum mechanics*, arXiv:2410.04262.
- [9] S. Laliberte, B. McPeak, *Bootstrapping supersymmetric (matrix) quantum mechanics*, arXiv:2510.01356.